Written HW3

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I use Neovim (BTW)

Full HW repository at
github.com/LoganCTanner/School

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1. Suppose that $X \sim normal(\mu, \sigma^2)$. Find the following probabilities:

   (a) $P\{|X-\mu| \leq \sigma\}$
   Answer:
   $P\{|X-\mu| \leq \sigma\} = P\{-\sigma \leq X-\mu \leq \sigma\}$
   $= P\{-1 \leq \frac{X-\mu}{\sigma} \leq 1\} = P\{\frac{X-\mu}{\sigma} \leq 1\} - \left(1 - P\{\frac{X-\mu}{\sigma} \leq 1\}\right)$
   $= 2P\{\frac{X-\mu}{\sigma} \leq 1\} - 1$
   Then, by the table of values provided for a standard normal curve,
   $= 1.6826 - 1 = \boxed{0.6826}$

   (b) $P\{|X-\mu| \leq 2\sigma\}$
   Answer:
   It follows from part (a) that
   $P\{|X-\mu| \leq 2\sigma\} = 2P\{\frac{X-\mu}{\sigma} \leq 2\} - 1 = \boxed{0.9544}$

   (c) $P\{|X-\mu| \leq 3\sigma\}$
   Answer:
   $P\{|X-\mu| \leq 3\sigma\} = 2P\{\frac{X-\mu}{\sigma} \leq 3\} - 1 = \boxed{0.9974}$

2. Some time ago, a random sample of 747 obituaries published in Salt Lake City newspapers found that 46% of the decedents died in the 3-month period following their birthdays.

   (a) What percent would you expect to die in that 3-month period? What assumption(s) are you making?
   Answer:
   Given the assumption that deaths are evenly distributed, it would follow that 25% of deaths would occur in the 3-month period following a birthday, given that would equate to 25% of the calendar year.

   (b) Estimate the probability that 46% or more of the decedents would die in the 3-month period following their birthdays.
   Answer:
   Given the assumption $X \sim \text{Binomial}(747, .25)$, it may be approximated through a standard normal distribution, where
   $46\% \times 747 = 343.62$
   $P(X \geq 343.62) = 1 - P(X < 343.62)$
   $E[X] = np = 186.75$
   $Var(X) = np(1-p) = 186.75(.75) = 140.0625$
   $\sigma = \sqrt{Var(X)} \approx 11.8348$
   Then, given that $\frac{X-\mu}{\sigma} = Z \sim \text{normal}(0,1)$,
   it follows that
   $$\frac{343.62 - 186.75}{11.8348} = Z \approx 13.25$$
   Since the Z value obtained is roughly 4 times larger than the largest Z value on the table provided, it can be concluded that
   $$P(X \geq 343.62) = 1 - P(X < 343.62) \approx 1 - 1 = \boxed{0}$$

   (c) What conclusion might you draw from your calculation in (b)? How many deaths in the 3-month period would you accept as consistent with your expectations?
   Answer:
   Conclusions may include:

      i. data collection or calculation error.
      ii. clustering of birthdays with some mass death event in the sampled decedent data.
      iii. underlying causal factor(s) in the mortality rate in the 3-month period following a birthday.

   In the expected even distribution case, it would be expected to have roughly 187 ± 12 deaths occur.

3. Suppose Y_j, where $1 \leq j \leq n$, are independent exponential random variables with parameters λ_j. Find the cdf and pdf for the random variable $M = \min(Y_1, Y_2, \ldots, Y_n)$. What type of random variable is M?
   Answer:
   Given the hint to find $P(M > m)$ for a given $M \geq 0$,

$$P(M > m) = P(Y_1 > m) \times \cdots \times P(Y_n > m)$$

$$P(Y_j > m) = 1 - P(Y_j < m) = 1 - F_Y(m) = e^{-\lambda m}$$

$$\Rightarrow P(M > m) = 1 - \prod_{i=1}^{n} e^{-\lambda_i m} = 1 - exp\left(-m \sum_{i=1}^{n} \lambda_i\right)$$

   Then, assigning $\sum \lambda_i = \lambda_M$ gives a cdf of,

$$P(M > m) = 1 - e^{-m\lambda_M}$$

   with the pdf given by

$$F'_M(m) = f_M(m) = 1 - e^{-m\lambda_M} \frac{d}{dm} = \lambda_M e^{-m\lambda_M}$$

   resulting in M being an exponential random variable, given its cdf and pdf match the general forms of each.

4. Suppose that an appliance is constructed in such a way that it requires that n independent electronic components are all functioning. Assume that the lifespan of each of the components, T_j, is an exponential random variable with parameter λ_j.

(a) Let X be the random variable giving the lifespan of the appliance. Find the distribution of X.
Answer:
As was obtained in question 3, lifespan X would be an exponential random variable, given that the appliance will break when one of its components break (or, in other words, the minimum T_j), and all components T_j are exponential random variables.

(b) Find $E[X]$. Then, find the **median** lifespan of the appliance, that is, the time t at which half of the appliances are likely to have broken and half still working.
Answer:
Given X is an exponential r. v., it follows that

$$E[X] = \frac{1}{\lambda_X}, \lambda_X = \sum_{i=1}^{n} \lambda_i$$

with the median being calculated by

$$F_X(t) = 1 - e^{-\lambda_X t} \Rightarrow \frac{1}{2} = 1 - e^{-\lambda_X t}$$

$$\frac{1}{2} = e^{-\lambda_X t} \Rightarrow t = -\ln\left(\frac{1}{2}\right)/\lambda_X = -\ln\left(\frac{1}{2}\right)E[X]$$

(c) Consider the two numbers you computed in (b). What does this indicate in context about the lifespans of these devices? Which will the manufacturer of the appliance likely use in their advertising?
Answer:
Given that $-\ln\left(\frac{1}{2}\right) \approx 0.693$, the average appliance will last longer than 50% of other such appliances. This would lead to the manufacturer likely advertising the average lifespan of the appliance.

5. Let $X \sim exponential(\lambda)$, and let $Y = e^X$.

(a) Find the cdf of Y.
Answer:

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(e^X \leq y) = P(X \leq \ln(y)) \\ &= 1 - e^{-\lambda \ln(y)} = \boxed{1 - y^{-\lambda}} \end{aligned}$$

(b) Use (a) to find the pdf of Y. Check that your answer really is a density by showing that it integrates to 1.

$$F'_Y(y) = f_Y(y) = 1 - y^{-\lambda} \frac{d}{dy} = \boxed{\lambda y^{-\lambda}}$$

Then, since the smallest possible value X can take is 0, then the smallest Y is $e^0 = 1$, giving limits of integration of $[1, \infty)$

$$\int_1^\infty \lambda y^{-\lambda-1} dy = -y^{-\lambda}\Big|_1^\infty = 0 - \left(-\frac{1}{1^\lambda}\right) = \boxed{1}$$

(c) Find the pdf of Y again, this time using the "shortcut" formula (the Theorem from Part 3 of the Ch 5 Notes), being sure to explain why it applies.